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Programme/Cohort	OCO/Cohort11	Date	19/09/2026
Instructor		Score	/100
Environment	Kali Linux / Ubuntu	Submission	E-Campus

LAB 1 REPORT ON CORE102

EVIDENCE:

```

kali@kali: ~/Documents/ICDFA_LABS/core102_lab1
Session Actions Edit View Help
└─$ cd core102_lab1

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
└─$ whoami
kali

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
└─$ hostname
kali

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
└─$ ip -br link
lo UNKNOWN 00:00:00:00:00:00 <LOOPBACK,UP,LOWER_UP>
eth0 UP 00:0c:29:f7:72:d2 <BROADCAST,MULTICAST,UP,LOWER_UP>

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
└─$ ip -br addr
lo UNKNOWN 127.0.0.1/8 ::1/128
eth0 UP 192.168.149.128/24 fe80::96d8:5426:4bfc:2254/64

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
└─$ ip route
default via 192.168.149.2 dev eth0 proto dhcp src 192.168.149.128 metric 100
192.168.149.0/24 dev eth0 proto kernel scope link src 192.168.149.128 metric 100

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
└─$ ip route get 8.8.8.8
8.8.8.8 via 192.168.149.2 dev eth0 src 192.168.149.128 uid 1000
cache

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]

```

RESPONSE:

Observation	Your result	Meaning
Active interface	Eth0	Interface used by the default route
IPv4 address and prefix	192.168.149.128/24	Host address and network prefix
Default gateway	default via 192.168.149.2	Router used for destinations outside the local subnet
Route source address	192.168.149.128	Address Linux selected for the test destination

LAB 2 REPORT

EVIDENCE:

DANJUMA YAHAYA ADDRA

IPv4 LAB REPORT

```

kali@kali: ~/Documents/ICDFA_LABS/core102_lab1
Session Actions Edit View Help
lo UNKNOWN 00:00:00:00:00:00 <LOOPBACK,UP,LOWER_UP>
eth0 UP 00:0c:29:f7:72:d2 <BROADCAST,MULTICAST,UP,LOWER_UP>

(kali@kali)~/Documents/ICDFA_LABS/core102_lab1
$ ip -br addr
lo UNKNOWN 127.0.0.1/8 ::1/128
eth0 UP 192.168.149.128/24 fe80::96d8:5426:4bfc:2254/64

(kali@kali)~/Documents/ICDFA_LABS/core102_lab1
$ ip route
default via 192.168.149.2 dev eth0 proto dhcp src 192.168.149.128 metric 100
192.168.149.0/24 dev eth0 proto kernel scope link src 192.168.149.128 metric 100

(kali@kali)~/Documents/ICDFA_LABS/core102_lab1
$ ip route get 8.8.8.8
8.8.8.8 via 192.168.149.2 dev eth0 src 192.168.149.128 uid 1000
cache

(kali@kali)~/Documents/ICDFA_LABS/core102_lab1
$ python3 - <<'PY'
for value in (192, 168, 10, 25):
    print(f'{value:3} = {value:08b}')
PY
192 = 11000000
168 = 10101000
10 = 00001010
25 = 00011001

(kali@kali)~/Documents/ICDFA_LABS/core102_lab1
$ python3 - <<'PY'

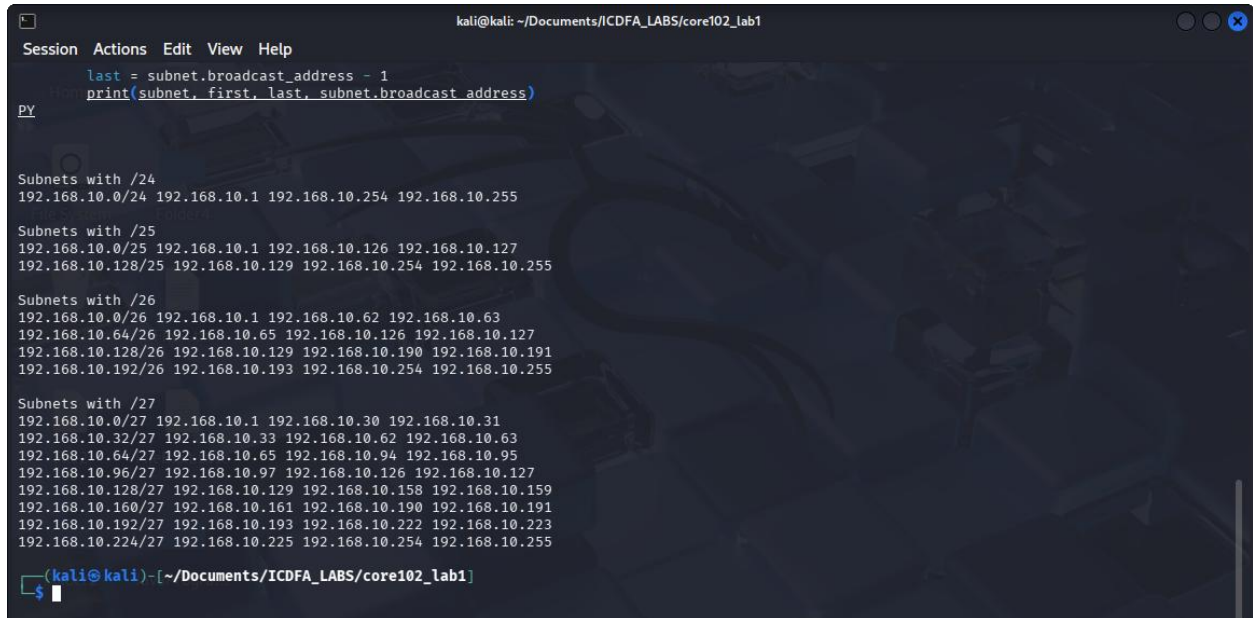
```

RESPONSE:

Address	Historical class	Given prefix	Network ID	Broadcast
10.24.6.9/20	A	/20	10.24.6.0	10.24.6.15
172.20.14.200/23	B	/23	172.20.14.0	172.20.15.255
192.168.44.77/27	C	/27	192.168.44.64	192.168.44.95
198.51.100.130/26	C	/26	198.51.100.128	198.51.100.191

LAB 3 REPORT

EVIDENCE:



```
kali@kali: ~/Documents/ICDFA_LABS/core102_lab1
Session Actions Edit View Help
last = subnet.broadcast_address - 1
print(subnet.first, last, subnet.broadcast_address)
PY

Subnets with /24
192.168.10.0/24 192.168.10.1 192.168.10.254 192.168.10.255

Subnets with /25
192.168.10.0/25 192.168.10.1 192.168.10.126 192.168.10.127
192.168.10.128/25 192.168.10.129 192.168.10.254 192.168.10.255

Subnets with /26
192.168.10.0/26 192.168.10.1 192.168.10.62 192.168.10.63
192.168.10.64/26 192.168.10.65 192.168.10.126 192.168.10.127
192.168.10.128/26 192.168.10.129 192.168.10.190 192.168.10.191
192.168.10.192/26 192.168.10.193 192.168.10.254 192.168.10.255

Subnets with /27
192.168.10.0/27 192.168.10.1 192.168.10.30 192.168.10.31
192.168.10.32/27 192.168.10.33 192.168.10.62 192.168.10.63
192.168.10.64/27 192.168.10.65 192.168.10.94 192.168.10.95
192.168.10.96/27 192.168.10.97 192.168.10.126 192.168.10.127
192.168.10.128/27 192.168.10.129 192.168.10.158 192.168.10.159
192.168.10.160/27 192.168.10.161 192.168.10.190 192.168.10.191
192.168.10.192/27 192.168.10.193 192.168.10.222 192.168.10.223
192.168.10.224/27 192.168.10.225 192.168.10.254 192.168.10.255

(kali@kali) - [~/Documents/ICDFA_LABS/core102_lab1]
```

RESPONSE: This lab teaches me how to subnet an ip address into two, four, and eight, equal addresses. I also used Python program to verify the division. I have understood the important role python plays in subnetting.

LAB 4 REPORT

EVIDENCE:

```
kali@kali: ~/Documents/ICDFA_LABS/core102_lab1
Session Actions Edit View Help
(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
$ ip link show core102lab0 2>/dev/null || echo "Name is available"
Name is available

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
$ sudo ip link add core102lab0 type dummy
[sudo] password for kali:

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
$ sudo ip link set core102lab0 up

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
$ sudo ip addr add 192.168.10.10/25 dev core102lab0

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
$ sudo ip addr add 192.168.10.130/25 dev core102lab0

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
$ ip -br addr show core102lab0
core102lab0      UNKNOWN      192.168.10.10/25 192.168.10.130/25 fe80::89f:21ff:fe13:143e/64

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
$ ping -c 2 192.168.10.10
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.
64 bytes from 192.168.10.10: icmp_seq=1 ttl=64 time=0.181 ms
64 bytes from 192.168.10.10: icmp_seq=2 ttl=64 time=0.123 ms

— 192.168.10.10 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1011ms
rtt min/avg/max/mdev = 0.123/0.152/0.181/0.029 ms

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
$ ping -c 2 192.168.10.10
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.
64 bytes from 192.168.10.10: icmp_seq=1 ttl=64 time=0.181 ms
64 bytes from 192.168.10.10: icmp_seq=2 ttl=64 time=0.123 ms

— 192.168.10.10 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1011ms
rtt min/avg/max/mdev = 0.123/0.152/0.181/0.029 ms

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
$ ping -c 2 192.168.10.130
PING 192.168.10.130 (192.168.10.130) 56(84) bytes of data.
64 bytes from 192.168.10.130: icmp_seq=1 ttl=64 time=0.028 ms
64 bytes from 192.168.10.130: icmp_seq=2 ttl=64 time=0.126 ms

— 192.168.10.130 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1015ms
rtt min/avg/max/mdev = 0.028/0.077/0.126/0.049 ms

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
$ sudo ip link delete core102lab0

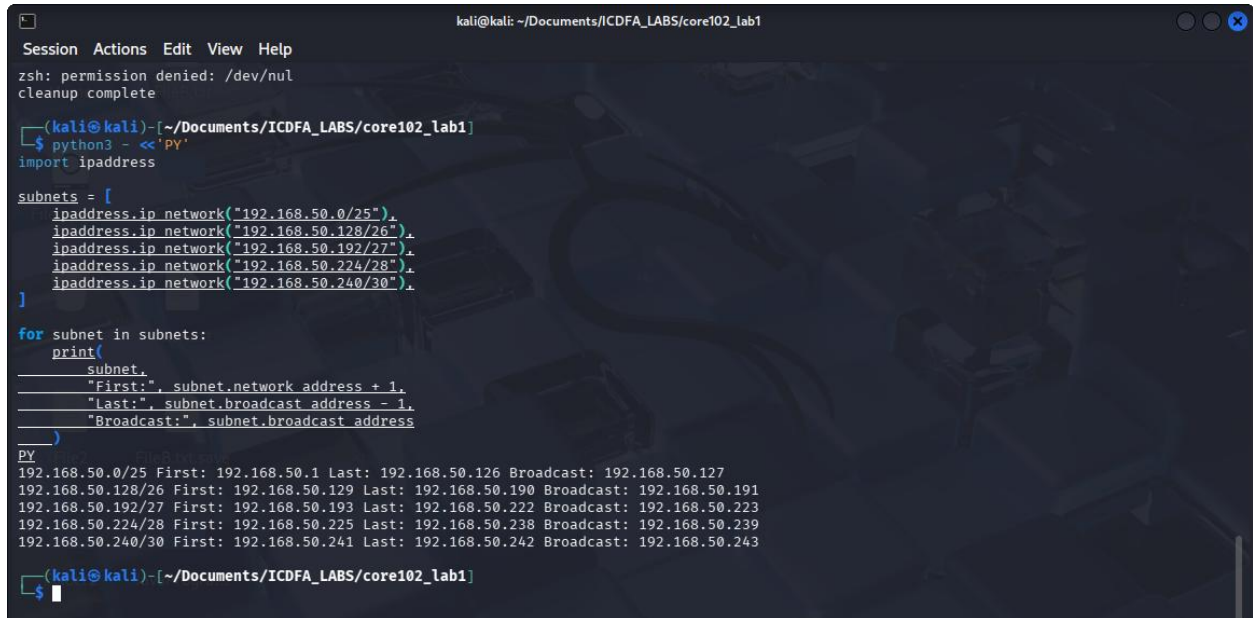
(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
$ sudo ip show core102lab0 2>/dev/nul || echo "cleanup complete"
zsh: permission denied: /dev/nul
cleanup complete

(kali@kali)~[~/Documents/ICDFA_LABS/core102_lab1]
$
```

RESPONSE: This Lab teaches me how to create and configure a temporary dummy interface which I can use for my practical without changing the kali network interface. I created and ran the configuration and also deleted the interface as recommended by the Lab activity.

LAB 5 REPORT

EVIDENCE:

A screenshot of a Kali Linux terminal window. The window title is 'kali@kali: ~/Documents/ICDFA_LABS/core102_lab1'. The terminal shows a message 'zsh: permission denied: /dev/nul' and 'cleanup complete'. The user is in a shell prompt '(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab1]'. They run 'python3 - <<'PY' import ipaddress'. Then they define a list 'subnets' with five entries: 'ipaddress.ip network("192.168.50.0/25")', 'ipaddress.ip network("192.168.50.128/26")', 'ipaddress.ip network("192.168.50.192/27")', 'ipaddress.ip network("192.168.50.224/28")', and 'ipaddress.ip network("192.168.50.240/30")'. They then loop through 'subnets' and print 'First:', 'Last:', and 'Broadcast:' addresses. The output shows the calculated ranges for each subnet. Finally, they run 'PY' to execute the script.

```
kali@kali: ~/Documents/ICDFA_LABS/core102_lab1
zsh: permission denied: /dev/nul
cleanup complete

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab1]
$ python3 - <<'PY'
import ipaddress

subnets = [
    ipaddress.ip network("192.168.50.0/25"),
    ipaddress.ip network("192.168.50.128/26"),
    ipaddress.ip network("192.168.50.192/27"),
    ipaddress.ip network("192.168.50.224/28"),
    ipaddress.ip network("192.168.50.240/30"),
]

for subnet in subnets:
    print(
        subnet,
        "First:", subnet.network address + 1,
        "Last:", subnet.broadcast address - 1,
        "Broadcast:", subnet.broadcast address
    )
PY
192.168.50.0/25 First: 192.168.50.1 Last: 192.168.50.126 Broadcast: 192.168.50.127
192.168.50.128/26 First: 192.168.50.129 Last: 192.168.50.190 Broadcast: 192.168.50.191
192.168.50.192/27 First: 192.168.50.193 Last: 192.168.50.222 Broadcast: 192.168.50.223
192.168.50.224/28 First: 192.168.50.225 Last: 192.168.50.238 Broadcast: 192.168.50.239
192.168.50.240/30 First: 192.168.50.241 Last: 192.168.50.242 Broadcast: 192.168.50.243

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab1]
$
```

RESPONSE: In this lab, I learned how to use VLSM (Variable Length Subnet Masking) to divide the network 192.168.50.0/24 into smaller networks based on the number of hosts required.

I allocated the largest subnet first and assigned:

Training Hall = /25

Staff Network = /26

Cyber Range = /27

Management = /28

Router Link = /30

I calculated the network ID, usable IP range, and broadcast address for each subnet and ensured that the subnets did not overlap. I also used Python's ipaddress module to verify my calculations.

LAB 6 REPORT

EVIDENCE:

```
kali@kali: ~/Documents/ICDFA_LABS/core102_lab1
Session Actions Edit View Help
192.168.50.240/30 First: 192.168.50.241 Last: 192.168.50.242 Broadcast: 192.168.50.243

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab1]
$ nano subnet_report.py

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab1]
$ chmod +x subnet_report.py

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab1]
$ ./subnet_report.py 192.168.10.173/27
File "/home/kali/Documents/ICDFA_LABS/core102_lab1/./subnet_report.py", line 28
    chmod +x subnet_report.py
    ^^^^^^^^^^^^^^^^^^^^^^^^^
SyntaxError: invalid syntax

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab1]
$ ./subnet_report.py
File "/home/kali/Documents/ICDFA_LABS/core102_lab1/./subnet_report.py", line 28
    chmod +x subnet_report.py
    ^^^^^^^^^^^^^^^^^^^^^^^^^
SyntaxError: invalid syntax

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab1]
$ nano subnet_report.py

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab1]
$ nano subnet_report.py

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab1]
$ ./subnet_report.py 192.168.10.173/27
```

```
kali@kali: ~/Documents/ICDFA_LABS/core102_lab1
Session Actions Edit View Help

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab1]
$ ./subnet_report.py 192.168.10.173/27

Host address      : 192.168.10.173
Prefix            : /27
Subnet mask       : 255.255.255.224
Network ID        : 192.168.10.160
Broadcast         : 192.168.10.191
Total addresses   : 32
First usable      : 192.168.10.161
Last usable       : 192.168.10.190
Traditional hosts : 30

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab1]
$ ./subnet_report.py 172.20.14.200/23

Host address      : 172.20.14.200
Prefix            : /23
Subnet mask       : 255.255.254.0
Network ID        : 172.20.14.0
Broadcast         : 172.20.15.255
Total addresses   : 512
First usable      : 172.20.14.1
Last usable       : 172.20.15.254
Traditional hosts : 510

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab1]
$ ./subnet_report.py 198.168.10.5
Usage: ./subnet_report.py ADDRESS/PREFIX

(kali@kali)-[~/Documents/ICDFA_LABS/core102_lab1]
$ ./subnet_report.py 192.168.10.5
```

```
kali@kali: ~/Documents/ICDFA_LABS/core102_lab1
Session Actions Edit View Help

(kali@kali)~/Documents/ICDFA_LABS/core102_lab1
$ ./subnet_report.py 198.168.10.5
Usage: ./subnet_report.py ADDRESS/PREFIX

(kali@kali)~/Documents/ICDFA_LABS/core102_lab1
$ ./subnet_report.py 192.168.10.5
Usage: ./subnet_report.py ADDRESS/PREFIX

(kali@kali)~/Documents/ICDFA_LABS/core102_lab1
$ ./subnet_report.py 192.168.10.5
Usage: ./subnet_report.py ADDRESS/PREFIX

(kali@kali)~/Documents/ICDFA_LABS/core102_lab1
$ nano subnet_report.py

(kali@kali)~/Documents/ICDFA_LABS/core102_lab1
$ ./subnet_report.py 192.168.10.5/22
Host address      : 192.168.10.5
Prefix            : /22
Subnet mask       : 255.255.252.0
Network ID        : 192.168.8.0
Broadcast         : 192.168.11.255
Total addresses   : 1024
First usable      : 192.168.8.1
Last usable       : 192.168.11.254
Traditional hosts : 1022

(kali@kali)~/Documents/ICDFA_LABS/core102_lab1
$ ./subnet_report.py 999.1.1.1/24
Invalid input: '999.1.1.1/24' does not appear to be an IPv4 or IPv6 interface
```

RESPONSE: In this lab, the Python program is used to calculate subnet information automatically.

The program showed the IP address, subnet mask, network ID, broadcast, and usable IP addresses.

I tested it with both valid and invalid IP addresses.

The lab helped me understand how Python can make subnetting easier and faster.

Overall, I learned how to use Python to verify my manual subnet calculations.